Robust Estimation of Graph Statistics

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1 Introduction

1.1 Background

Nowadays, many datasets can be naturally represented as graphs, and it is often of interest to study the underlying structure of these graphs. One important example is the problem of community detection, where we want to find clusters of nodes that are more densely connected to each other than to the rest of the graph. The stochastic block model (SBM) is a popular random graph model that captures this community structure, and has been widely studied in the literature. It randomly assigns each vertex to one of k communities, and then generates edges independently between vertices with probabilities that depend on their community memberships. This model is simple yet rich enough to capture many real-world phenomena, and has been used as a benchmark for testing community detection. In this paper, we will focus on community detection under the balanced, two-community SBM, which we define below.

Definition 1.1 (Balanced, two-community SBM)**.** *A graph $G \sim \mathcal{G}(n, p, q)$ is a balanced, two-community stochastic block model with $V = [n]$ and uniformly random label vector $\boldsymbol{x} \in \{\pm 1\}^n$ satisfying $\mathbf{1}^\top \boldsymbol{x} = 0$ if, for each pair of vertices $i, j \in [n]$, we have*

$$\mathbb{P}((i,j) \in E) = \begin{cases} p & \text{if } x_i = x_j, \\ q & \text{if } x_i \neq x_j. \end{cases}$$

Usually, we have $p > q$, so that vertices in the same community are more likely to be connected than those in different communities, demonstrating the community structure.

In practice, however, real data never perfectly follows any theoretical model. Randomness or corruption in the data can lead to nontrivial deviations, so it is important to design algorithms that are robust to such perturbations. In the context of community detection, this means that we want to develop algorithms that can still recover the underlying community structure in some capacity, even under these contaminations. Thus, in this essay, we will study algorithms that can achieve robust recovery of the community labels under various perturbation models. In particular, we focus on algorithms based on semidefinite programming (SDP), which has been used to tackle a wide range of robustness problems.

1.2 Evaluating recovery

In our setting, we can measure the recovery quality by the correlation between the true community labels and the estimated community labels. Suppose $\hat{\boldsymbol{x}}$ is the estimated community label vector, then we can define the correlation as the inner product $\langle \hat{\boldsymbol{x}}, \boldsymbol{x} \rangle / n$. If we guess the labels randomly, then the expected correlation is zero. We define the idea of recovery quality as follows, following [Abb23].

Definition 1.2 (Recovery quality)**.** *Given a balanced, two-community SBM G with true community labels $\boldsymbol{x}$, an estimated community label vector $\hat{\boldsymbol{x}}$ achieves:*

- **exact recovery,** *if $|\langle \hat{\boldsymbol{x}}, \boldsymbol{x} \rangle| / n = 1$ with high probability;*

- **almost exact recovery,** *if* $|\langle \hat{\boldsymbol{x}}, \boldsymbol{x} \rangle|/n = 1 - o(1)$ *with high probability;*

- **weak recovery,** *if* $|\langle \hat{\boldsymbol{x}}, \boldsymbol{x} \rangle|/n > \alpha$ *for some* $\alpha > 0$ *with high probability.*

In this paper, when we say "with high probability", we mean that the probability goes to 1 as the number of vertices n goes to infinity.

1.3 Semidefinite programming

Semidefinite programming (SDP) is a powerful tool in optimization. It is an extension of linear programming where we optimize a linear function over the set of positive semidefinite matrices satisfying some set of linear constraints. Formally, an SDP can be written in the following standard form:

$$\begin{aligned} \text{minimize} \quad & \langle \boldsymbol{C}, \boldsymbol{X} \rangle \\ \text{subject to} \quad & \langle \boldsymbol{A}_i, \boldsymbol{X} \rangle = b_i \text{ for } i = 1, \ldots, m, \\ & \boldsymbol{X} \succeq 0, \end{aligned}$$

where $\langle \cdot, \cdot \rangle$ denotes the matrix inner product (sum of entrywise products), $\boldsymbol{C}, \boldsymbol{A}_1, \ldots, \boldsymbol{A}_m$ are given symmetric matrices, $b_1, \ldots, b_m$ are given scalars, and $\boldsymbol{X}$ is the variable matrix we want to optimize over.

In our context, we will usually have $\boldsymbol{C} = \boldsymbol{A}_G - \boldsymbol{K}$, where $\boldsymbol{A}_G$ is the adjacency matrix of the graph G, and $\boldsymbol{K}$ is some correction matrix that depends on the model parameters and the observed graph. The constraints will typically enforce that $\boldsymbol{X}$ has diagonal entries all equal to 1, and sometimes other linear constraints. We will use the following notations.

Definition 1.3 (Semidefinite programming notations)**.** *Denote the set of positive semidefinite matrices with diagonal entries equal to* 1 *by*

$$\mathcal{PSD}_1(n) = \{\boldsymbol{X} \in \mathbb{R}^{n \times n} : \boldsymbol{X} \succeq 0, \boldsymbol{X}_{ii} = 1 \text{ for all } i \in [n]\}.$$

Write $\mathcal{A}$ *for the set of matrices satisfying some linear constraints, i.e.*

$$\mathcal{A} = \{\boldsymbol{X} \in \mathbb{R}^{n \times n} : \langle \boldsymbol{A}_i, \boldsymbol{X} \rangle = b_i \text{ for } i = 1, \ldots, m\}.$$

Given a matrix $\boldsymbol{M} \in \mathbb{R}^{n \times n}$*, we define*

$$\mathcal{SDP}_{\mathcal{A}}(\boldsymbol{M}) := \max_{\boldsymbol{X} \in \mathcal{PSD}_1(n) \cap \mathcal{A}} \langle \boldsymbol{M}, \boldsymbol{X} \rangle.$$

In particular, if there are no extra constraints, we omit the subscript $\mathcal{A}$ *and simply write* $\mathcal{SDP}(\boldsymbol{M})$*.*

1.4 Structure of this essay

The rest of the essay is structured as follows.

In Section 2, we review some existing robustness models in community detection, and the corresponding SDP-based algorithms.

In Section 3, we focus on the main problem of this essay, which is to achieve weak recovery at the Kesten-Stigum threshold under adversarial edge perturbations, explained in details in Section 2.2. We will take a detailed look at why the algorithm by Montanari and Sen [MS16] works, and how it is automatically robust against edge perturbations.

In Section 4, we introduce a new perturbation model, where we allow the adversary to modify the edge probabilities directly. We claim that the same algorithm by Montanari and Sen remains robust, and we briefly compare this notion of robustness with other existing notions discussed in Section 2.

In Section 5, we prove the robustness claim above.

Finally, in Section 6, we conclude the essay and discuss some future directions.

2 Existing robustness models in community detection

2.1 Monotone adversary, exact recovery

One of the earlier works on robust community detection is by Feige and Kilian [FK01] in 2001. They considered the following monotone adversary model, and came up with an SDP algorithm that achieves exact recovery against this adversary.

check Boppana

Definition 2.1 (Monotone adversary model)**.** *Let G be a balanced, two-community SBM. A monotone adversary can look at the realised graph G and the community labels x of the vertices, and they are allowed to make the following changes:*

(a) Add edges between vertices in the same community.

(b) Remove edges between vertices in different communities.

It may seem that the monotone adversary can only make the problem easier, since they are either: (a), strengthening the community structure; or (b), removing inter-community connections. However, certain naive algorithms already fail under this model.

2.1.1 Naive example: common neighbors algorithm

Consider the following naive algorithm taking in two nodes, and outputting whether they are in the same community or not.

Algorithm 1 (Common neighbors algorithm)**.**

0. Take inputs: graph G on n vertices; $i, j \in [n]$: the two vertices to investigate; d: threshold for degree comparison.

1. Count the number of common neighbors n_{ij} of i and j.

2. Output "same community" if $n_{ij}/n \geq r(p,q)$, and "different communities" otherwise.

The $r(p,q)$ threshold is a constant that depends on p and q. The idea behind the algorithm is that, for i and j in the same community, the expected number of common neighbors is

$$\left(\frac{n}{2}-2\right)p^2 + \frac{n}{2}q^2 \approx \frac{p^2+q^2}{2}n,$$

where the first term counts their common neighbors in their community, and the second term counts their common neighbors in the other community. On the other hand, for i and j in different communities, the expected number of common neighbors is

$$(n-2)pq \approx (pq)n.$$

Since $p^2+q^2 > 2pq$ for $p > q$, we can choose $r(p,q) \in (pq, (p^2+q^2)/2)$, so the algorithm works with high probability.

Clearly, the algorithm above relies on the exact values of p and q heavily. The monotone adversary can easily break this. For example, it can add all the edges in both communities. Then, for any two vertices in different communities, their expected number of common neighbors is $(n-2)q \approx qn$. To combat this, we might want to instead choose $r(p,q) \in (q, (p^2+q^2)/2)$, but this can be impossible, as q can be larger than $(p^2+q^2)/2$ when p and q are small, which is the sparse regime we are interested in. Thus, the common neighbors algorithm is not robust to the monotone adversary.

2.1.2 SDP-based robust algorithm

Now, we consider the following SDP problem. Consider the following linear constrained set:

$$\mathcal{A} = \{\boldsymbol{X} \in \mathbb{R}^{n\times n} : \sum_{i,j\in[n]} \boldsymbol{X}_{ij} = 0\}.$$

Then, our SDP of interest is $\tilde{h}(G) = \mathcal{SDP}_{\mathcal{A}}(\boldsymbol{A}_G)$, which can be written explicitly as

$$\begin{aligned} \tilde{h}(G) = \text{maximize} \quad & \langle \boldsymbol{A}_G, \boldsymbol{X}\rangle \\ \text{subject to} \quad & \boldsymbol{X} \succeq 0, \\ & \boldsymbol{X}_{ii} = 1 \text{ for all } i \in [n], \\ & \sum_{i,j\in[n]} \boldsymbol{X}_{ij} = 0. \end{aligned}$$

To gain some intuition behind this SDP, let us consider a set of balanced community labels $\hat{x} \in \{\pm 1\}^n$. It is easy to see that the matrix $\hat{\boldsymbol{X}} := \hat{x}\hat{x}^\top$ is a feasible solution to the above SDP. Plugging in $\hat{\boldsymbol{X}}$ into the

objective, we have

$$\begin{aligned}
\langle \boldsymbol{A}_G, \hat{\boldsymbol{X}} \rangle &= \sum_{i,j \in [n]} \boldsymbol{A}_{G,ij} \hat{x}_i \hat{x}_j = \sum_{i,j \in E} \hat{x}_i \hat{x}_j \\
&= \sum_{i,j \text{ in same community}} 1 + \sum_{i,j \text{ in different communities}} (-1) \\
&= 2 \times \text{number of edges within communities} - 2 \times \text{number of edges between communities} \\
&= 2|E| - 4 \times \text{number of edges between communities.}
\end{aligned}$$

Thus, if we maximize the above SDP over all balanced community labels, we are essentially trying to minimize the number of edges between communities. By making the SDP relaxation, we make the problem tractable. Writing $h(G) := |E|/2 - \tilde{h}(G)/4$, $b(G) :=$ number of edges between communities, then we see that $h(G) \leq b(G)$ is an SDP lower bound on the number of edges between communities. In fact, we often have $h(G) = b(G)$, by the following result.

Lemma 2.1 (Lemma 19 of [FK01]). *Let G be a balanced, two-community SBM with edge probabilities p and q as defined above. If $p - q \geq C\sqrt{p \log n / n}$ for a sufficiently large constant C, then with probability at least $1 - 2n^{-5}$, we have $h(G) = b(G)$.*

Furthermore, it can be shown that modifying one edge of G preserves the equality $h(G) = b(G)$ with high probability.

Lemma 2.2 (Lemma 20 of [FK01]). *Let G be a graph satisfying $h(G) = b(G)$. Let G' be the graph obtained by modifying one edge of G (adding a new edge or removing an existing one). Then, with probability at least $1 - n^{-1+o(1)}$, we have $h(G') = b(G')$ for all possible modifications.*

This inspires the following algorithm.

cite FK, Boppana

Algorithm 2 (SDP-based exact recovery under monotone adversary).

0. Take input: graph G on n vertices.
1. For $i \in [n] \backslash \{1\}$, let G_i be the graph obtained by "toggling" the edge $(1, i)$: i.e. if G contains the edge $(1, i)$, remove it; otherwise, add it.
2. Writing $S_1 = \{1\} \cup \{i \in [n] \backslash \{1\} : h(G_i) = h(G)\}$, output the community labels $\hat{\boldsymbol{x}}$ as follows:

$$\hat{\boldsymbol{x}}_i = \begin{cases} 1 & \text{if } i \in S_1, \\ -1 & \text{if } i \notin S_1. \end{cases}$$

We can show that this algorithm recovers the true community labels with high probability, and is robust to any monotone adversary.

Theorem 2.1 (Exact recovery under monotone adversary, Theorem 5 of [FK01]). *Let G be a balanced, two-community SBM with intra-community edge probability p and inter-community edge probability q, and*

C be a sufficiently large constant. If $p - q \geq C\sqrt{p \log n / n}$, then with high probability, the above algorithm recovers the true community labels, and is robust to any monotone adversary.

Proof. We work under the event that $h(G) = b(G)$, and that for all $i \in [n] \backslash \{1\}$, $h(G_i) = b(G_i)$. By Lemmas 2.1 and 2.2, this event occurs with high probability.

If 1 and i are in the same community, then the edge $(1, i)$ is an intra-community edge, and will not affect the inter-community edge count b, so we have $b(G_i) = b(G)$. On the other hand, if 1 and i are in different communities, then the edge $(1, i)$ is indeed an inter-community edge, so $b(G_i) = b(G) \pm 1$, so $h(G_i) \neq h(G)$. Hence, we've shown exact recovery.

To show robustness against the monotone adversary, it suffices to show the following:

Lemma 2.3 (Monotone modifications preserve $h = b$)**.** *Let G be a balanced SBM satisfying $h(G) = b(G)$. Let G' be the graph obtained by making one monotone modification to G (i.e. adding an edge between vertices in the same community, or removing an edge between vertices in different communities). Then, we have $h(G') = b(G')$.*

Note that this lemma is an exact statement, so we can make arbitrary many monotone modifications. Combining this lemma with Lemmas 2.1 and 2.2, we have that $h(G) = b(G)$, and $h(G_i) = b(G_i)$ for all $i \in [n] \backslash \{1\}$ still holds with high probability after any number of monotone modifications, and thus the algorithm still works.

To prove this lemma, the following three properties of h will be helpful:

- **Lower bound of b:** Since $b(G)$ is the value of the objective at a feasible solution to the SDP, we have

$$h(G) \leq b(G).$$

- **Increasing:** Writing G^+ for the graph obtained by adding an edge (u, v) to G, and $\boldsymbol{X}^+$, $\boldsymbol{X}$ for the optimal solutions to the SDP for G^+ and G respectively, we have

$$\begin{aligned} h(G^+) - h(G) &= \left(\frac{|E(G^+)|}{2} - \frac{\tilde{h}(G^+)}{4} \right) - \left(\frac{|E(G)|}{2} - \frac{\tilde{h}(G)}{4} \right) \\ &= \frac{1}{2} - \frac{1}{4} \left(\langle \boldsymbol{A}_{G^+}, \boldsymbol{X}^+ \rangle - \langle \boldsymbol{A}_G, \boldsymbol{X} \rangle \right) \\ &\geq \frac{1}{2} - \frac{1}{4} \left(\langle \boldsymbol{A}_{G^+}, \boldsymbol{X} \rangle - \langle \boldsymbol{A}_G, \boldsymbol{X} \rangle \right) \\ &= \frac{1}{2} - \frac{1}{4} (\boldsymbol{X}_{uv} + \boldsymbol{X}_{vu}) \geq 0, \end{aligned}$$

where we used the fact that $|\boldsymbol{X}|_\infty \leq 1$. So $h(G^+) \geq h(G)$.

- **Bounded increments:** Proceeding in exactly the same way as above, we have

$$\begin{aligned} h(G^+) - h(G) &= \frac{1}{2} - \frac{1}{4}\left(\langle \boldsymbol{A}_{G^+}, \boldsymbol{X}^+\rangle - \langle \boldsymbol{A}_G, \boldsymbol{X}\rangle\right) \\ &\leq \frac{1}{2} - \frac{1}{4}\left(\langle \boldsymbol{A}_{G^+}, \boldsymbol{X}^+\rangle - \langle \boldsymbol{A}_G, \boldsymbol{X}^+\rangle\right) \\ &= \frac{1}{2} - \frac{1}{4}(\boldsymbol{X}_{uv} + \boldsymbol{X}_{vu}) \leq 1, \end{aligned}$$

where we again used the fact that $|\boldsymbol{X}|_\infty \leq 1$. So $h(G^+) \leq h(G) + 1$.

Now, we are ready to prove the lemma. We call the bisection of G that achieves the optimal value $b(G)$ the *optimal bisection* of G, and the two communities created by this bisection the *optimal communities* of G.

If G^+ is obtained by adding an edge between vertices in the same community, the number of edges between any two balanced communities can only increase. In particular, the number of edges between the optimal communities of G remains the same, so this bisection remains optimal for G^+. Thus, $b(G^+) = b(G)$. By the lower bound property, we have $h(G^+) \leq b(G^+) = b(G) = h(G)$, and by the increasing property, we have $h(G^+) \geq h(G)$. Hence, all the inequalities are equalities, and we have $h(G^+) = b(G^+)$.

On the other hand, if G^- is obtained by removing an edge between vertices in different communities, the number of edges between any two balanced communities can at most decrease by one. In particular, the number of edges between optimal communities of G also decreases by one, so this bisection remains optimal for G^-. Thus, $b(G^-) = b(G) - 1$. By the lower bound property, we have $h(G^-) \leq b(G^-) = b(G) - 1 = h(G) - 1$, and by the bounded increments property, we have $h(G^-) \geq h(G) - 1$. Hence, all the inequalities are equalities, and we have $h(G^-) = b(G^-)$.

Therefore, $h(G) = b(G)$ is preserved under any monotone modification, which proves the lemma, and thus the theorem. □

Remark 2.1. Even though monotone adversaries are quite limited in their power, the above result is still interesting in that it does not require any knowledge of the parameters p and q. Furthermore, it highlights the fundamental reason why SDP-based algorithms can be robust: they are increasing, but have bounded increments, so small modifications to the graph does not change the output by much. In this particular case, by leveraging the fact that the optimal objective value exactly correlates with an important graph statistic (the minimal number of edges between communities) with high probability, we are able to strengthen the robustness guarantee to hold for any number of monotone modifications. In our cases, we will not be able to achieve such a strong guarantee, but this "bounded difference" property of SDP will still be crucial. For example, in Section 2.2, we will relate $\mathcal{SDP}(\boldsymbol{A}_G^{\mathsf{cent}})$ to the SDP value of some Gaussian matrix, which will be the central component allowing the robust algorithm to work.

2.2 Edge-robust weak recovery

Now let us consider a different setting, where we want to instead achieve weak recovery with the minimal amount of community structure. We first introduce the following formulation of the SBM, which is more

convenient for our analysis.

Definition 2.2 (d, ϵ-SBM formulation)**.** *A graph $G \sim \mathcal{G}(n, d, \epsilon)$ is a balanced, two-community stochastic block model graph with $V = [n]$ and uniformly random label vector $\boldsymbol{x} \in \{\pm 1\}^n$ satisfying $\mathbf{1}^\top \boldsymbol{x} = 0$, if for each pair of vertices $i, j \in [n]$, we have*

$$\mathbb{P}((i,j) \in E) = \begin{cases} \frac{d}{n}(1 + \frac{\epsilon}{2}) & \text{if } x_i = x_j, \\ \frac{d}{n}(1 - \frac{\epsilon}{2}) & \text{if } x_i \neq x_j. \end{cases}$$

Note that d is the average degree of the graph, and ϵ controls the strength of the community structure. These parameters are assumed to be known constants, and we are interested in the case where n could be very large. Hence, we are working in a sparse regime compared to the previous section, where p and q are constants.

How much community structure do we really need to achieve weak recovery? It has been shown that [MNS15], in the sparse regime where d is constant, weak recovery is possible if and only if

$$\lambda := \frac{\epsilon\sqrt{d}}{2} > 1.$$

This threshold is known as the Kesten-Stigum threshold. There are various algorithms that can achieve weak recovery above the Kesten-Stigum threshold, but they are not known to be robust against edge perturbations. In [MS16], Montanari and Sen proposed a simple SDP-based algorithm that can achieve weak recovery almost at the Kesten-Stigum threshold, and is automatically robust against edge perturbations.

Let us first introduce the notion of the centered adjacency matrix.

Definition 2.3 (Centered adjacency matrix)**.** *Given a graph G, let $\boldsymbol{A}_G$ be its adjacency matrix, and define $\boldsymbol{A}_G^{\text{cent}} = \boldsymbol{A}_G - \mathbb{E}[\boldsymbol{A}_G]$. For $G \sim \mathcal{G}(n, d, \epsilon)$, we have $\mathbb{E}[\boldsymbol{A}_G] = \frac{d}{n}\mathbf{1}\mathbf{1}^\top$.*

In their paper, Montanari and Sen looked at $\mathcal{SDP}(\boldsymbol{A}_G^{\text{cent}})$, the basic SDP problem on the centered adjacency matrix, and proposed the following algorithm.

Algorithm 3 (SDP Algorithm from [MS16])**.**

1. Sample edges of G independently with probability $(1 + n^{-1/2})^{-1}$ to obtain the sampled edge set $E_{\text{sample}} \subset E$, and then compute the evaluation edge set $E_{\text{eval}} = E \setminus E_{\text{sample}}$.
2. Construct the subgraphs $G_{\text{sample}} = (V, E_{\text{sample}})$ and $G_{\text{eval}} = (V, E_{\text{eval}})$.
3. Compute an optimiser $\boldsymbol{X}^*$ of $\mathcal{SDP}(\boldsymbol{A}_{G_{\text{sample}}}^{\text{cent}})$, i.e., $\langle \boldsymbol{A}_{G_{\text{sample}}}^{\text{cent}}, \boldsymbol{X}^* \rangle = \mathcal{SDP}(\boldsymbol{A}_{G_{\text{sample}}}^{\text{cent}})$.
4. Compute the eigenvectors $\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(n)}$ of $\boldsymbol{X}^*$.

5. For each $i, j \in [n]$, define $\hat{\boldsymbol{x}}^{(i,j)} \in \{-1, 0, 1\}^n$ as follows:

$$\hat{\boldsymbol{x}}_k^{(i,j)} = \begin{cases} \operatorname{sign}(\boldsymbol{v}_k^{(i)}) & \text{if } |\boldsymbol{v}_k^{(i)}| \geq |\boldsymbol{v}_j^{(i)}|, \\ 0 & \text{otherwise.} \end{cases}$$

6. Find $(i^*, j^*) = \arg\max_{i,j \in [n]} (\hat{\boldsymbol{x}}^{(i,j)})^\top \boldsymbol{A}_{G_{\text{eval}}}^{\text{cent}} \hat{\boldsymbol{x}}^{(i,j)} = \sum_{a,b \in E_{\text{eval}}} \hat{\boldsymbol{x}}_a^{(i,j)} \cdot \hat{\boldsymbol{x}}_b^{(i,j)}$.
7. Output $\hat{\boldsymbol{x}}^{\text{SDP}}(G) = \hat{\boldsymbol{x}}^{(i^*,j^*)}$.

We have the following weak recovery rate and edge robustness result for Algorithm 3.

Theorem 2.2 (Weak Recovery for Algorithm 3)**.** *Let $G \sim \mathcal{G}(n, d, \epsilon)$ and assume, for some $\delta > 0$, that $\lambda = \epsilon\sqrt{d}/2 \geq 1 + \delta$. Then, there exists $\eta = \eta(\delta) > 0$ and $d_* = d_*(\delta) > 0$ such that, for all $d \geq d_*$, the output of Algorithm 3, $\hat{\boldsymbol{x}}^{\text{SDP}}(G)$, satisfies*

$$\mathbb{P}\left(\frac{1}{n}|\langle \hat{\boldsymbol{x}}^{\text{SDP}}(G), x\rangle| \geq \eta\right) \geq 1 - Ce^{-\sqrt{n}/C}.$$

This will be the first main result we will focus on later in this essay in Section 3. For now, we will just give a high-level overview of the algorithm and the proof ideas.

The main idea behind the proof is that there exists some Gaussian random matrix $\boldsymbol{B}(\lambda)$, such that $\mathcal{SDP}(\boldsymbol{A}_G^{\text{cent}}/\sqrt{d})$ is close to $\mathcal{SDP}(\boldsymbol{B}(\lambda))$, and $\mathcal{SDP}(\boldsymbol{A}_G^{\text{cent}}/\sqrt{d} - (\lambda/n)\boldsymbol{x}\boldsymbol{x}^\top)$ is close to $\mathcal{SDP}(\boldsymbol{B}(0))$. There is a phase transition result for the SDP value of $\boldsymbol{B}(\lambda)$ at $\lambda = 1$, so we can bound $\mathcal{SDP}(\boldsymbol{A}_G^{\text{cent}}/\sqrt{d})$ and $\mathcal{SDP}(\boldsymbol{A}_G^{\text{cent}}/\sqrt{d} - (\lambda/n)\boldsymbol{x}\boldsymbol{x}^\top)$ apart. This is known as the *push-out effect*, which guarantees that the optimizer to $\mathcal{SDP}(\boldsymbol{A}_G^{\text{cent}}/\sqrt{d})$ must contain nontrivial information about $\boldsymbol{x}$. Then, we can show that the rounding procedure in Step 5 and the selection procedure in Step 6 can extract this information to produce an estimator $\hat{\boldsymbol{x}}^{\text{SDP}}(G)$ that achieves weak recovery.

If we allow a small number of edge perturbations, then by the bounded differences property of SDP, the two SDP values, $\mathcal{SDP}(\boldsymbol{A}_G^{\text{cent}}/\sqrt{d})$ and $\mathcal{SDP}(\boldsymbol{A}_G^{\text{cent}}/\sqrt{d} - (\lambda/n)\boldsymbol{x}\boldsymbol{x}^\top)$, only undergo small changes and remain bounded apart, so the rest of the proof still goes through.

Remark 2.2. Note that in the proof here, we again leveraged the "bounded differences" property of SDP, explicitly in the edge robustness part, and also implicitly in the node robustness part where we relate SDP values of the observed graph to those of some Gaussian random matrices. Then, using an additional SDP phase transition result for the Gaussian random matrices, we get the desired recovery and robustness guarantees. This mirrors the proof structure of the exact recovery under monotone adversary before, where we also leveraged the "bounded differences" property of SDP, and a different property of the SDP value to get recovery and robustness guarantees.

2.3 From edge-robust to node-robust

As is, the algorithm above is only known to be robust against $O(n\sqrt{d})$ edge perturbations. However, if we want to edit μn nodes, we can potentially change $O(n^2)$ edges, since editing each node can change up to n edges. Thus, the algorithm above is not automatically robust against node perturbations.

Later, Ding et al. [DdHS23] came up with a way to modify the algorithm above to make it robust against node perturbations. Their starting point is that, after μn node edits, there still remains a $(1-\mu)n \times (1-\mu)n$ submatrix of the adjacency matrix that is unperturbed, and thus still has the push-out effect. In particular, writing the set of unperturbed nodes as S, then $\boldsymbol{A}_G^{\text{cent}}{}_{[S,S]}$ has the push-out effect.

Thus, we want to select such a submatrix, a naive idea is to select the $(1-\mu)n \times (1-\mu)n$ submatrix with the largest SDP value, but this could be fabricated by the adversary. Instead, they propose to select the $(1-\mu)n \times (1-\mu)n$ submatrix whose large submatrices all have large SDP values. This is a more robust selection procedure, and

2.4 Degree heterogeneity

So far, we have only considered SBMs where all nodes have the same expected degree. In practice, this is not always the case. We can consider the more general degree-corrected SBM (DCSBM) model, where each node is given an additional expected degree parameter, and the edge probabilities are determined by both the community labels and the expected degree parameters of the endpoints.

Definition 2.4 (Balance, two-community, degree-corrected SBM)**.** *A graph $G \sim \mathcal{G}(n, p, q, \boldsymbol{\theta})$ is a degree-corrected SBM graph with $V = [n]$, and uniformly random label vector $\boldsymbol{x} \in \{\pm 1\}^n$ satisfying $\mathbf{1}^\top \boldsymbol{x} = 0$, if, for each pair of vertices $i, j \in [n]$, we have*

$$\mathbb{P}((i,j) \in E) = \begin{cases} \theta_i \theta_j p & \text{if } x_i = x_j, \\ \theta_i \theta_j q & \text{if } x_i \neq x_j. \end{cases}$$

This definition can be extended to multiple unbalanced communities, but for consistency with previous sections, we will only consider the balanced, two-community case here. The results we will discuss are originally proven in the more general setting, but the balanced, two-community case is already sufficient to illustrate the main ideas.

2.5 Outlier nodes

3 Proof of weak recovery (Theorem 2.2)

restructure and move main theorem to this section

Before we start the proof, we will make a few definitions, and show a few preliminary results. Let $\boldsymbol{x}$ be the label vector of $G \sim \mathcal{G}(n, d, \epsilon)$. Note that $G_{\text{sample}} \sim \mathcal{G}(n, d_{\text{sample}}, \epsilon)$, and $G_{\text{eval}} \sim \mathcal{G}(n, d_{\text{eval}}, \epsilon)$, where $d_{\text{sample}} = d/(1 + n^{-1/2})$ and $d_{\text{eval}} = n^{-1/2} d_{\text{sample}}$. For simplicity, from now on we write $d = d_{\text{sample}}$.

Throughout the proof, we will use the constant C to denote a positive constant that changes from line to line. We will often use the fact that, if two events both hold with probability at least $1 - Ce^{-n/C}$, then their intersection also holds with probability at least $1 - Ce^{-n/C}$, by a simple union bound, and

$$C_1 e^{-n/C_1} + C_2 e^{-n/C_2} \leq (C_1 + C_2) e^{-n/\max(C_1, C_2)} \leq (C_1 + C_2) e^{-n/(C_1+C_2)} = Ce^{-n/C}.$$

Also, note that, in our definition, G_{sample} and G_{eval} are dependent. However, we need their independence in our proof. To fix this issue, we note that if we sampled them independently, then the two graphs

investigate independence

3.1 Step 1: $\boldsymbol{X}^*$ contains nontrivial information about $\boldsymbol{x}$

We first introduce a group of Gaussian random matrices (GOEs) that have well-behaved SDP values.

Definition 3.1 (Gaussian orthogonal ensemble (GOE) and the deformed GOE $\boldsymbol{B}(\lambda)$)**.** *A random matrix $\boldsymbol{W} \in \mathbb{R}^{n \times n}$ is a* GOE matrix *if it is a symmetric matrix with $\{\boldsymbol{W}_{ij}\}_{i \leq j}$ independent and $\boldsymbol{W}_{ij} \sim \mathcal{N}(0, 1/n)$ for $i < j$ and $\boldsymbol{W}_{ii} \sim \mathcal{N}(0, 2/n)$. The* deformed GOE matrix $\boldsymbol{B}(\lambda)$ *is then defined as*

$$\boldsymbol{B}(\lambda) = \frac{\lambda}{n} \mathbf{1}\mathbf{1}^\top + \boldsymbol{W}.$$

We then have the following phase transition result for the SDP value of $\boldsymbol{B}(\lambda)$.

Lemma 3.1 (SDP value of $\boldsymbol{B}(\lambda)$, Theorem 5 of [MS16])**.** *Let $\boldsymbol{B}(\lambda)$ be a deformed GOE matrix defined in Definition 3.1. Then, with probability converging to 1 as $n \to \infty$,*

(a) if $\lambda \leq 1$: for any $\epsilon > 0$, we have $\mathcal{SDP}(\boldsymbol{B}(\lambda))/n \in [2 - \epsilon, 2 + \epsilon]$.

(b) if $\lambda > 1$: there exists $\Delta(\lambda) > 0$ such that $\mathcal{SDP}(\boldsymbol{B}(\lambda))/n \geq 2 + \Delta(\lambda)$.

If we set the diagonal entries of $\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}$ to be $\lambda\sqrt{d}/n$, then

$$\frac{1}{\sqrt{d}} \boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}} = \frac{\lambda}{n} \boldsymbol{x}\boldsymbol{x}^\top + \boldsymbol{E},$$

where $\boldsymbol{E}$ is a symmetric matrix with zero diagonal and independent off-diagonal entries satisfying

$$\boldsymbol{E}_{ij} = \begin{cases} \frac{1}{\sqrt{d}} - \frac{\sqrt{d}}{n} - \frac{\lambda}{n} \boldsymbol{x}_i \boldsymbol{x}_j & \text{with probability } \mathbb{P}((i,j) \in E_{\text{sample}}), \\ -\frac{\sqrt{d}}{n} - \frac{\lambda}{n} \boldsymbol{x}_i \boldsymbol{x}_j & \text{with probability } 1 - \mathbb{P}((i,j) \in E_{\text{sample}}). \end{cases}$$

Let $\boldsymbol{X}^*$ be an optimizer of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}})$. We will now try to bound $\langle \boldsymbol{E}, \boldsymbol{X}^* \rangle$ and $\langle \frac{1}{\sqrt{d}} \boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \rangle$ away from each other using the phase transition result of Lemma 3.1, which will then give us a lower bound on $n^{-2} \langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \rangle$, which is the correlation between $\boldsymbol{X}^*$ and the ground truth labels matrix $\boldsymbol{x}\boldsymbol{x}^\top$.

The following lemma bounds $\mathcal{SDP}(\boldsymbol{E})$ to the SDP value of a GOE matrix.

Lemma 3.2 (GOE approximation of $\mathcal{SDP}(\boldsymbol{E})$)**.** *Let $\boldsymbol{W}$ be a GOE matrix. Then, there exists $C = C(\lambda) > 0$*

such that, for all $n \geq n_0(d, \epsilon)$ with probability at least $1 - Ce^{-n/C}$, we have

$$\left| \frac{1}{n}\mathcal{SDP}(\boldsymbol{E}) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{W}) \right| \leq \frac{C\log(d)}{d^{1/10}}.$$

Moreover, $C(\lambda)$ is bounded for λ in any compact set.

We also have a similar result bounding $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}/\sqrt{d})$ by the SDP value of a deformed GOE matrix.

Lemma 3.3 (Deformed GOE approximation of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}/\sqrt{d})$)**.** *Let $\boldsymbol{B}(\lambda)$ be a deformed GOE matrix. Then, there exists $C = C(\lambda) > 0$ such that, for all $n \geq n_0(d, \epsilon)$*

$$\left| \frac{1}{n}\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}\right) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{B}(\lambda)) \right| \leq \frac{C\log(d)}{d^{1/10}}.$$

Moreover, $C(\lambda)$ is bounded for λ in any compact set.

We will discuss the proofs of the above two lemmas later in Section 5. We will take them for granted here.

Now we have the tools to obtain a lower bound on $n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^*\rangle$.

$$\begin{aligned}
\left\langle \frac{\lambda}{n^2}\boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \right\rangle &= \frac{1}{n}\left(\left\langle \frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \right\rangle - \langle \boldsymbol{E}, \boldsymbol{X}^*\rangle \right) \\
&= \frac{1}{n}\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}\right) - \frac{1}{n}\langle \boldsymbol{E}, \boldsymbol{X}^*\rangle \\
&\geq \frac{1}{n}\mathcal{SDP}(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{E}) && \text{by optimality of SDP} \\
&\geq \frac{1}{n}\mathcal{SDP}(\boldsymbol{B}(\lambda)) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{W}) - \frac{C\log(d)}{d^{1/10}} && \text{by Lemmas 3.2 and 3.3} \\
&\geq \Delta_1(\lambda) - \frac{C\log(d)}{d^{1/10}}, && \text{by Lemma 3.1}
\end{aligned}$$

with probability at least $1 - Cn^{-4}$ (see the remark below).

Remark 3.1. In the original paper [MS16], the authors claimed that the above inequality holds with probability at least $1 - Ce^{-n/C}$, but tracking the proof of Lemma 3.1 (Appendices F and G of [MS16]) shows that they only proved the result with probability at least $1 - Cn^{-4}$ (Lemma G.8 of [MS16]). This is still sufficient for our purposes, as the probability still tends to 1 as $n \to \infty$.

Now fix λ, and take d sufficiently large such that $\Delta_1(\lambda) - \frac{C\log(d)}{d^{1/10}} \geq \lambda\Delta_2(\lambda) > 0$. Then, we have, with probability at least $1 - Cn^{-4}$,

$$n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^*\rangle \geq \Delta_2(\lambda).$$

This is an important milestone, as it shows that $\boldsymbol{X}^*$ has a nontrivial correlation with the ground truth $\boldsymbol{x}\boldsymbol{x}^\top$.

3.2 Step 2: There exists "good" label estimators among $\{\hat{\boldsymbol{x}}^{(i,j)}\}_{i,j\in[n]}$

We first define the following rounding function used in step 5 of Algorithm 3:

Definition 3.2 (Rounding function). *For $t \geq 0$, define the* rounding function $r_t : \mathbb{R} \to \{-1, 0, +1\}$ *as follows:*

$$r_t(x) = \begin{cases} 1 & \text{if } x \geq t, \\ 0 & \text{if } -t < x < t, \\ -1 & \text{if } x \leq -t. \end{cases}$$

Hence, we can write $\hat{\boldsymbol{x}}^{(\hat{\imath},j)} = r_{\boldsymbol{v}_j^{(\hat{\imath})}}(\boldsymbol{v}^{(\hat{\imath})})$.

Since $\boldsymbol{X} \in \mathcal{PSD}_1(n)$, writing its eigen-decomposition as $\boldsymbol{X} = \sum_{i=1}^n \xi_i \boldsymbol{v}^{(i)}(\boldsymbol{v}^{(i)})^\top$, where ξ_i is the i-th largest eigenvalue, and $\boldsymbol{v}^{(i)}$ is the corresponding unit length eigenvector, we have $\xi_i \geq 0$ for all i by positive semidefiniteness, and $\sum_{i=1}^n \xi_i = \mathrm{Tr}(\boldsymbol{X}) = n$. Thus,

$$\Delta_2(\lambda) \leq \frac{1}{n^2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \rangle = \frac{1}{n^2}\sum_{i=1}^n \xi_i (\boldsymbol{x}^\top \boldsymbol{v}^{(i)})^2 = \sum_{i=1}^n \frac{\xi_i}{n} \cdot \left(\frac{\boldsymbol{x}^\top \boldsymbol{v}^{(i)}}{\sqrt{n}}\right)^2.$$

So there exists $\hat{\imath} \in [n]$ such that $|\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})}| \geq \sqrt{n\Delta_2(\lambda)}$.

Next, we will use following lemma to justify that the rounding function preserves this correlation, which shows that it is possible to extract a good label estimator.

Lemma 3.4 (Rounding Lemma). *Let X, Y be random variables such that: $\mathbb{P}(X = +1) = \mathbb{P}(X = -1) = 1/2$, $\mathbb{E}(XY) \geq \epsilon > 0$, and $\mathbb{E}(Y^2) = 1$. Then, there exists t^* depending on the joint distribution of X and Y, such that*

$$\mathbb{E}(X r_{t^*}(Y)) \geq \frac{\epsilon^2}{2}.$$

Proof. Note that for all t, we have $X r_t(Y) = r_t(XY)$, and $Y^2 = (XY)^2$. Let $Z = XY$, then it suffices to show that given $\mathbb{E}(Z) \geq \epsilon > 0$ and $\mathbb{E}(Z^2) = 1$, we have $\mathbb{E}(r_t(Z)) \geq \epsilon^2/2$. Recall that for nonnegative random variable S, by Fubini, we have

$$\mathbb{E}(S) = \mathbb{E}\left(\int_0^\infty \mathbb{1}_{s \leq S} ds\right) = \int_0^\infty \mathbb{P}(S \geq s) ds, \quad \mathbb{E}(S^2) = \mathbb{E}\left(\int_0^\infty 2s \mathbb{1}_{s \leq S} ds\right) = \int_0^\infty 2s\mathbb{P}(S \geq s) ds.$$

Therefore, writing $Z^\pm = \max(\pm Z, 0)$, we have

$$\begin{aligned} \epsilon \leq \mathbb{E}(Z) = \mathbb{E}(Z_+) - \mathbb{E}(Z_-) &= \int_0^\infty \mathbb{P}(Z_+ \geq z) dz - \int_0^\infty \mathbb{P}(Z_- \geq z) dz \\ &\leq \int_0^K (\mathbb{P}(Z \geq z) - \mathbb{P}(Z \leq -z)) dz + \int_K^\infty \frac{2z}{2K}\mathbb{P}(Z \geq z) dz - \int_K^\infty \frac{2z}{2K}\mathbb{P}(-Z \geq z) dz \\ &= \int_0^K \mathbb{E} r_t(Z) dt + \frac{1}{2K}\mathbb{E}(Z_+^2 - Z_-^2) = \int_0^K \mathbb{E} r_t(Z) dt + \frac{1}{2K}. \end{aligned}$$

Now take $K = 1/\epsilon$, then

$$\frac{1}{K}\int_0^K \mathbb{E} r_t(Z) dt \geq \frac{1}{K}(\epsilon - \frac{1}{2K}) = \frac{\epsilon^2}{2}$$

So the average of $\mathbb{E}r_t(Z)$ over $t \in [0, 1/\epsilon]$ is greater than $\epsilon^2/2$, so there exists $t^* \in [0, 1/\epsilon]$ such that $\mathbb{E}r_{t^*}(Z) \geq \epsilon^2/2$. $\square$

Without loss of generality, assume that $\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})} > 0$. Consider (X, Y) drawn uniformly from $\{(\boldsymbol{x}_k, \sqrt{n}\boldsymbol{v}_k^{(\hat{\imath})})\}_{k\in[n]}$, then we have $\mathbb{E}(XY) = n^{-1/2}\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})} \geq \sqrt{\Delta_2(\lambda)}$ and $\mathbb{E}(Y^2) = n^{-1}||\sqrt{n}\boldsymbol{v}^{(\hat{\imath})}||^2 = ||\boldsymbol{v}^{(\hat{\imath})}||^2 = 1$. Thus, by Lemma 3.4, there exists $t^* \in [0, 1/\sqrt{\Delta_2(\lambda)}]$ such that

$$\frac{\Delta_2(\lambda)}{2} \leq \mathbb{E}(Xr_{t^*}(Y)) = \frac{1}{n}\sum_{k=1}^{n} \boldsymbol{x}_k r_{t^*}(\sqrt{n}\boldsymbol{v}_k^{(\hat{\imath})}) = \frac{1}{n}\langle \boldsymbol{x}, r_{\tilde{t}}(\boldsymbol{v}^{(\hat{\imath})})\rangle,$$

where $\tilde{t} = t^*/\sqrt{n} \in [0, 1/\sqrt{n\Delta_2(\lambda)}]$.

Remark 3.2. Here, we slightly improved the rounding lemma bound from $\epsilon^2/4$ in [MS16] to $\epsilon^2/2$, although the main proof idea remains the same. Also, the original application of the rounding lemma missed the scaling factor $\sqrt{n}$ in the definition of Y, but the conclusion remains valid.

Finally, note that for any value of $\tilde{t}$, we can find $\hat{\jmath} \in [n]$ such that $r_{\tilde{t}}(\boldsymbol{v}^{(\hat{\imath})}) = r_{\boldsymbol{v}_{\hat{\jmath}}^{(\hat{\imath})}}(\boldsymbol{v}^{(\hat{\imath})}) = \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})}$. Therefore, there exists some $\hat{\imath}, \hat{\jmath} \in [n]$ such that

$$\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})}\rangle \geq \frac{\Delta_2(\lambda)}{2}. \tag{3.1}$$

Now, it remains to show that the selection step (Step 6) of Algorithm 3 will find such $\hat{\imath}, \hat{\jmath}$ with high probability, which will then complete the proof of Theorem 2.2.

3.3 Step 3: A "good" estimator can be found with G_{eval}

We first want to relate the quantities $\hat{\boldsymbol{x}}^\top \boldsymbol{A}_{G_{\text{eval}}}^{\text{cent}} \hat{\boldsymbol{x}}$, the objective function in Step 6 of Algorithm 3, to $\langle \boldsymbol{x}, \hat{\boldsymbol{x}}\rangle$, the correlation above, with high probability. Recall the well-known Azuma-Hoeffding inequality.

add reference

Lemma 3.5 (Azuma-Hoeffding inequality)**.** *Let $X_0, \ldots, X_m$ be a martingale, with bounded increments, i.e. $|X_i - X_{i-1}| \leq c_i$ for all $i \in [m]$. Then, for all $t > 0$, we have*

$$\mathbb{P}(|X_m - \mathbb{E}(X_m)| \geq t) \leq 2\exp\left(-\frac{2t^2}{\sum_{i=1}^m c_i^2}\right).$$

We apply this inequality to the objective function in Step 6 of Algorithm 3, and obtain the following concentration result.

Lemma 3.6. *Let $G \sim \mathcal{G}(n, d, \epsilon)$. For any fixed vector $\hat{\boldsymbol{x}} \in \{-1, 0, +1\}^n$, define $f(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G^{\text{cent}} \hat{\boldsymbol{x}}$. Then, for any $t > 0$,*

$$\mathbb{P}\Big(\big|f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}}))\big| \geq t \;\Big|\; |E_{\text{eval}}| \leq m\Big) \leq 2\exp\left(-\frac{t^2}{8m}\right).$$

Proof. Note that $f(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G^{\text{cent}} \hat{\boldsymbol{x}} = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G \hat{\boldsymbol{x}} - (d/n)(\hat{\boldsymbol{x}}^\top \mathbf{1})^2$, so writing $\tilde{f}(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G \hat{\boldsymbol{x}}$, we have $f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}})) = \tilde{f}(G; \hat{\boldsymbol{x}}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}))$. Note that $\tilde{f}(G; \hat{\boldsymbol{x}}) = \sum_{(a,b)\in E} 2\hat{\boldsymbol{x}}_a \hat{\boldsymbol{x}}_b$.

For some ordering of all possible edges, we can define a filtration $\mathcal{F}_0, \mathcal{F}_1, \ldots, \mathcal{F}_m$ with $\mathcal{F}_k = \sigma(E_1, \ldots, E_k)$, where $E_k \in E(G) \cup \varnothing$ is the k-th edge of G in the ordering. Then, define the martingale $X_k = \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | \mathcal{F}_k)$ for $k \in [m]$. Since $|E(G)| \leq m$, we have $X_m = \tilde{f}(G; \hat{\boldsymbol{x}})$. Note that for all $k \in [m]$, we have

$$\begin{aligned}
|X_k - X_{k-1}| &= \left| \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1}) \right| \\
&\leq \sup_{e_{1:k-1}, e_k, e_k'} \left| \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1} = e_{1:k-1}, E_k = e_k) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1} = e_{1:k-1}, E_k = e_k') \right| \\
&\leq \sup_{E(G^+) = E(G) \cup \{(a,b)\}} \left| \tilde{f}(G^+; \hat{\boldsymbol{x}}) - \tilde{f}(G; \hat{\boldsymbol{x}}) \right| \\
&\leq \sup_{a,b} |2 \hat{\boldsymbol{x}}_a \hat{\boldsymbol{x}}_b| \leq 2.
\end{aligned}$$

Hence, we can apply the Azuma-Hoeffding inequality Lemma 3.5 to X with $c_i = 2$ for all $i \in [m]$ to obtain

$$\begin{aligned}
\mathbb{P}\Big(\big| f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}})) \big| \geq t \;\Big|\; |E(G)| \leq m \Big) &= \mathbb{P}\Big(\big| \tilde{f}(G; \hat{\boldsymbol{x}}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}})) \big| \geq t \;\Big|\; |E(G)| \leq m \Big) \\
&= \mathbb{P}(|X_m - \mathbb{E}(X_m)| \geq t) \\
&\leq 2 \exp\left(-\frac{t^2}{8m} \right).
\end{aligned}$$

as desired. □

To apply the above concentration result, we need to bound $|E_{\text{eval}}| \sim \text{Binomial}\left(|E|, n^{-1/2}/(1 + n^{-1/2})\right)$, which we can achieve with the following Chernoff bound argument.

Lemma 3.7 (Bounding $|E_{\text{eval}}|$)**.** *Let $p_n = n^{-1/2}/(1 + n^{-1/2})$. Then, for some constant $C > 0$, we have*

$$\mathbb{P}(|E_{\text{eval}}| \geq C\sqrt{n}) \leq C e^{-\sqrt{n}/C}.$$

Proof. Recall that $|E| \sim \text{Binomial}\left(\binom{n}{2}, d/n\right)$, so by Chernoff bound, for all $s > 0$, we have

$$\begin{aligned}
\mathbb{P}(|E| \geq K_0 n) &\leq e^{-sK_0 n} \mathbb{E}(e^{s|E|}) = e^{-sK_0 n} \left(1 - \frac{d}{n} + \frac{d}{n} e^s \right)^{\binom{n}{2}} \\
&\leq \exp\left(-sK_0 n + \frac{d}{n}(e^s - 1)\binom{n}{2} \right) \\
&\leq \exp\left(\left(\frac{d}{2}(e^s - 1) - sK_0 \right) n \right) \\
&\leq C_0 e^{-n/C_0},
\end{aligned}$$

where the last inequality holds when K_0 is sufficiently large, and C_0 is a constant depending on d.

Now, on the event $\{|E| \leq K_0 n\}$ with probability at least $1 - C_0 e^{-n/C_0}$, we have $|E_{\text{eval}}| \sim \text{Binomial}\left(|E|, p_n\right)$,

so by Chernoff bound again, for all $s > 0$, we have

$$\begin{aligned}
\mathbb{P}(|E_{\text{eval}}| \geq K_1\sqrt{n}) &= \mathbb{E}\left(\mathbb{P}(|E_{\text{eval}}| \geq K_1\sqrt{n} \mid |E|)\right) \leq e^{-sK_1\sqrt{n}}\mathbb{E}(\mathbb{E}(e^{s|E_{\text{eval}}|} \mid |E|)) \\
&= e^{-sK_1\sqrt{n}}\mathbb{E}((1-p_n+p_ne^s)^{|E|}) \leq e^{-sK_1\sqrt{n}}\left(1+p_n(e^s-1)\right)^{K_0n} \\
&= \exp\left(-sK_1\sqrt{n}+K_0(e^s-1)np_n\right) \leq \exp\left((K_0(e^s-1)-sK_1)\sqrt{n}\right) \\
&\leq C_1e^{-\sqrt{n}/C_1},
\end{aligned}$$

where the last inequality holds when K_1 is sufficiently large, and C_1 is a constant depending on d. Combining the two probability bounds, we have

$$\mathbb{P}(|E_{\text{eval}}| \geq K\sqrt{n}) \leq C_0e^{-n/C_0} + C_1e^{-\sqrt{n}/C_1} \leq Ce^{-\sqrt{n}/C},$$

for suitable constants C, as desired. □

expand the following argument slightly

By Lemmas 3.6 and 3.7, for all $s \in [0,1]$, we have, with probability at least $1 - Ce^{-\sqrt{n}s^2/C}$,

$$\left|\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\hat{\boldsymbol{x}} - \mathbb{E}\left(\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\hat{\boldsymbol{x}}\right)\right| \leq s\sqrt{n}.$$

Note that

$$\begin{aligned}
\mathbb{E}\left(\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\hat{\boldsymbol{x}}\right) &= \hat{\boldsymbol{x}}^\top \mathbb{E}\left(\boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\right)\hat{\boldsymbol{x}} \\
&= \hat{\boldsymbol{x}}^\top \frac{\epsilon d}{2n}\left(\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{I}_n\right)\hat{\boldsymbol{x}} \\
&= \frac{\epsilon d}{2n}(\hat{\boldsymbol{x}}^\top\boldsymbol{x})^2 - \frac{\epsilon d}{2}.
\end{aligned}$$

Thus, with probability at least $1 - Ce^{-\sqrt{n}s^2/C}$, we have

$$\left|\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\hat{\boldsymbol{x}} - \frac{\epsilon d}{2n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}\rangle^2 + \frac{\epsilon d}{2}\right| \leq s\sqrt{n}. \tag{3.2}$$

This concentration result relates $\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\hat{\boldsymbol{x}}$ to the correlation $\langle \boldsymbol{x}, \hat{\boldsymbol{x}}\rangle$ with high probability.

In particular, at the end of the previous step (Equation (3.1)), we have shown that there exists some $\hat{\imath}, \hat{\jmath} \in [n]$ such that $\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})}\rangle \geq \frac{\Delta_2(\lambda)}{2}$. Then, by the concentration result 3.2, with probability at least $1 - Ce^{-\sqrt{n}/C}$ (taking $s = 1$), we have

$$\frac{1}{n}\hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})\top}\boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})} \geq \frac{\epsilon d}{2}\left(\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})}\rangle\right)^2 - \frac{\epsilon d}{2n} - \frac{1}{\sqrt{n}} \geq \frac{\epsilon d}{8}\Delta_2(\lambda)^2 - \frac{\epsilon d}{2n} - \frac{1}{\sqrt{n}} \geq \frac{\epsilon d}{12}\Delta_2(\lambda)^2 =: \Delta_3(\lambda),$$

where the last inequality follows from

$$\frac{\epsilon d}{2n} + \frac{1}{\sqrt{n}} \leq \frac{\epsilon d}{24}\Delta_2(\lambda)^2,$$

which holds for sufficiently large n. Therefore,

$$\max_{i,j\in[n]} \frac{1}{n}\hat{\boldsymbol{x}}^{(i,j)\top}\boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\hat{\boldsymbol{x}}^{(i,j)} \geq \frac{1}{n}\hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})\top}\boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})} \geq \Delta_3(\lambda).$$

On the other hand, suppose that $\hat{\boldsymbol{x}}$ has a lower correlation with $\boldsymbol{x}$, say $\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}\rangle \leq \frac{\Delta_2(\lambda)}{4}$, then by the concentration result 3.2, with probability at least $1 - Ce^{-\sqrt{n}/C}$, we have

$$\frac{1}{n}\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\hat{\boldsymbol{x}} \leq \frac{\epsilon d}{2}\left(\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}\rangle\right)^2 + \frac{\epsilon d}{2n} + \frac{1}{\sqrt{n}} \leq \frac{\epsilon d}{32}\Delta_2(\lambda)^2 + \frac{\epsilon d}{24}\Delta_2(\lambda)^2 < \frac{\epsilon d}{12}\Delta_2(\lambda)^2 = \Delta_3(\lambda).$$

Thus, with probability at least $1-Ce^{-\sqrt{n}/C}$, the maximizer $\hat{\boldsymbol{x}}^{\text{SDP}} = \hat{\boldsymbol{x}}^{(i^*,j^*)}$ of $\max_{i,j\in[n]} \hat{\boldsymbol{x}}^{(i,j)\top}\boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}}\hat{\boldsymbol{x}}^{(i,j)}$ must satisfy

$$\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{\text{SDP}}\rangle \geq \frac{\Delta_2(\lambda)}{4}.$$

This completes the proof of Theorem 2.2. ■

3.4 Robustness against modifying $O(n\sqrt{d})$ edges

4 Robustness of the SDP

explain

4.1 Robustness with perturbed SBMs

We first define the following more general class of random graph models, the *perturbed SBM*. We will then show that the same algorithm still works for this more general class of random graph models, which demonstrates the robustness of the algorithm and the SDP approach in general.

Definition 4.1 (Perturbed SBM)**.** *A graph $G \sim \mathcal{G}(n,d,\epsilon)$ is a balanced, two-community h-perturbed stochastic block model graph with $V=[n]$, uniformly random label vector $\boldsymbol{x}\in\{\pm1\}^n$ satisfying $\mathbf{1}^\top\boldsymbol{x}=0$, i.i.d. latent variables $\xi_i \sim \text{Unif}([0,1])$ for $i\in[n]$, and perturbation function $h:[0,1]^2\to\mathbb{R}$ satisfying $\mathbb{E}[h(\xi_i,\xi_j)]=0$ for all $i,j\in[n]$, if for each pair of vertices $i,j\in[n]$, we have*

$$p_{ij} := \mathbb{P}((i,j)\in E) = \begin{cases} \frac{d}{n}(1+\frac{\epsilon}{2}+h(\xi_i,\xi_j)) & \text{if } x_i = x_j, \\ \frac{d}{n}(1-\frac{\epsilon}{2}+h(\xi_i,\xi_j)) & \text{if } x_i \neq x_j. \end{cases}$$

We write $\eta_{ij} = x_ix_j\epsilon/2 + h(\xi_i,\xi_j)$, and $h_{ij}=h(\xi_i,\xi_j)$ for short.

We have the following weak recovery result for the perturbed SBM.

Theorem 4.1 (Weak recovery for perturbed SBM)**.** *Let $G\sim\mathcal{G}(n,d,\epsilon)$ be a balanced, two-community h-perturbed SBM graph, and let $\lambda=\epsilon^2 d/4$. Suppose there exists $\alpha>0$, such that:*

(a) the perturbation h satisfies that $\|h\|_\infty \leq Cd^{-2\alpha}$,

(b) let $\boldsymbol{H}$ be the matrix with entries $H_{ij}=h(\xi_i,\xi_j)$, then $\mathbb{E}(\|\boldsymbol{H}\|_{op}) \leq Cnd^{-1/2-\alpha}$.

If $\lambda > 1$, then there exists some constant $\Delta_4(\lambda) > 0$, such that with probability at least $1 - Cn^{-4}$, the estimator $\hat{\boldsymbol{x}}^{\mathrm{SDP}}$ returned by Algorithm 3 satisfies

check the probability bound

$$\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{\mathrm{SDP}} \rangle \geq \Delta_4(\lambda).$$

Remark 4.1. We can view the theorem above as a different notion of robustness of the algorithm, where instead of directly editing edges, we are allowed to perturb the edge probabilities by some random $h(\xi_i, \xi_j)$. The theorem shows that as long as the perturbation is not too large or too structured, the algorithm can still achieve weak recovery.

4.2 Comparison of the two notions of robustness

Note that the perturbation can lead to more than $O(n\sqrt{d})$ edge modifications in expectation.

Roughly speaking, the operator norm bound on $\boldsymbol{H}$ in condition (b) of Theorem 4.1 controls the degree of structure in the perturbation. For example, if $h(\xi_i, \xi_j)$'s are i.i.d., then $\mathbb{E}\|\boldsymbol{H}\|_{\mathsf{op}} \sim O(\sqrt{n})$, and since $d \ll n$, this bound is easily satisfied, with individual edge perturbations of order $O(d^{-2\alpha})$ to satisfy the L_∞ bound in condition (a). On the other hand, if $h(\xi_i, \xi_j)$'s are highly structured, say $\boldsymbol{H} = \boldsymbol{u}\boldsymbol{u}^\top$ for some vector $\boldsymbol{u} \in \mathbb{R}^n$, then $\|\boldsymbol{H}\|_{\mathsf{op}} = O(n)$, so the individual edge perturbations must be of order $O(d^{-1/2-\alpha})$.

flesh out the above argument with more citations

concrete example with more than $O(n\sqrt{d})$ edge modifications

5 Proof of weak recovery for perturbed SBMs (Theorem 4.1)

Note that the proof of weak recovery in Section 3 only uses the adjacency matrix of the graph in two places: the SDP approximation lemmas (Lemmas 3.2 and 3.3), which we will discuss for the rest of this section, and $\mathbb{E}[\boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}]$ in Equation (3.2), which will be the same for the perturbed SBM and the original SBM since $\mathbb{E}[h(\xi_i, \xi_j)] = 0$ for all $i, j \in [n]$. Thus, to show that the algorithm remains effective for the perturbed SBM, we only need to show that similar SDP approximation results hold for the perturbed SBM.

Therefore, in this section, we will demonstrate how to get a similar SDP approximation result as in Lemmas 3.2 and 3.3, which will prove theorem 4.1. The proofs of the lemmas will follow the exact same route, so they are ommitted here for brevity. We refer the reader to the original paper [MS16] for more details.

5.1 Helper functions

The main idea of approximating the SDP of adjacency matrices with the SDP of deformed GOE matrices is to utilize the following two helper functions:

Definition 5.1 ($\mathcal{OPT}_k$)**.** *Define the rank-constrained SDP value for $\boldsymbol{M} \in \mathbb{R}^{n\times n}$ and $k \in [n]$:*

$$\mathcal{OPT}_k(\boldsymbol{M}) = \max_{\substack{\boldsymbol{X} \in \mathcal{PSD}_1(n) \\ \mathrm{rank}(\boldsymbol{X}) \leq k}} \langle \boldsymbol{M}, \boldsymbol{X} \rangle.$$

Equivalently, let $\mathbb{S}^{k-1} = \{\boldsymbol{v} \in \mathbb{R}^k : ||\boldsymbol{v}||_2 = 1\}$ be the unit sphere in $\mathbb{R}^k$, then we have

$$\mathcal{OPT}_k(\boldsymbol{M}) = \max_{\boldsymbol{\sigma}_1,\ldots,\boldsymbol{\sigma}_n \in \mathbb{S}^{k-1}} \sum_{i,j=1}^{n} M_{ij}\langle \boldsymbol{\sigma}_i, \boldsymbol{\sigma}_j\rangle.$$

Definition 5.2 (Smoothed OPT)**.** *For $\beta > 0$, define $\Phi(\beta, k; \boldsymbol{M})$, the smoothed version of $\mathcal{OPT}_k(\boldsymbol{M})$, as follows:*

$$\Phi(\beta, k; \boldsymbol{M}) = \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta \sum_{i,j=1}^{n} M_{ij}\langle \boldsymbol{\sigma}_i, \boldsymbol{\sigma}_j\rangle\right) d\nu(\boldsymbol{\sigma})\right),$$

where ν is the uniform probability measure on $(\mathbb{S}^{k-1})^n$.

We can see that as $\beta \to \infty$, $\Phi(\beta, k; \boldsymbol{M})$ will select the maximum value of $\sum_{i,j=1}^{n} M_{ij}\langle \boldsymbol{\sigma}_i, \boldsymbol{\sigma}_j\rangle$, which is exactly $\mathcal{OPT}_k(\boldsymbol{M})$.

In their paper, Montanari and Sen showed the following bounds relating $\mathcal{SDP}(\boldsymbol{M})$, $\mathcal{OPT}_k(\boldsymbol{M})$ and $\Phi(\beta, k; \boldsymbol{M})$. We will quote these results without proof.

Theorem 5.1 (SDP vs OPT, Theorem 4 Equation 15 of [MS16])**.** *For $k \geq k_0$, where k_0 is a sufficiently large constant, for any symmetric matrix $\boldsymbol{M} \in \mathbb{R}^{n\times n}$, we have*

$$\left(1 - \frac{1}{k}\right)\mathcal{SDP}(\boldsymbol{M}) - \frac{1}{k}|\mathcal{SDP}(-\boldsymbol{M})| \leq \mathcal{OPT}_k(\boldsymbol{M}) \leq \mathcal{SDP}(\boldsymbol{M}).$$

Remark 5.1. Using just the first inequality, we have

$$\mathcal{OPT}_k(\boldsymbol{M}) + \mathcal{OPT}_k(-\boldsymbol{M}) \geq \left(1 - \frac{2}{k}\right)\left(|\mathcal{SDP}(\boldsymbol{M})| + |\mathcal{SDP}(-\boldsymbol{M})|\right),$$

so subtract by $\mathcal{SDP}(\boldsymbol{M})$ on every term and flipping signs, we get

$$0 \leq \mathcal{SDP}(\boldsymbol{M}) - \mathcal{OPT}_k(\boldsymbol{M}) \leq \frac{1}{k}\left(|\mathcal{SDP}(\boldsymbol{M})| + |\mathcal{SDP}(-\boldsymbol{M})|\right) \leq \frac{1}{k-2}\left(\mathcal{OPT}_k(\boldsymbol{M}) + \mathcal{OPT}_k(-\boldsymbol{M})\right).$$

Remark 5.2. Clearly, $\boldsymbol{A}_G^{\text{cen}}$ and $\boldsymbol{B}(\lambda)$ are symmetric matrices. Montanari and Sen showed that with probability at least $1 - Ce^{-n/C}$, we have (Theorem 8 equation 47 of [MS16])

$$\max\left\{\frac{1}{n\sqrt{d}}\mathcal{OPT}_k(\pm\boldsymbol{A}_G^{\text{cen}}), \frac{1}{n}\mathcal{OPT}_k(\pm\boldsymbol{B}(\lambda))\right\} \leq C,$$

so in fact, for $\boldsymbol{M} = \boldsymbol{A}_G^{\text{cen}}$ or $\boldsymbol{B}(\lambda)$, with probability at least $1 - Ce^{-n/C}$, we have

$$|\mathcal{SDP}(\boldsymbol{M}) - \mathcal{OPT}_k(\boldsymbol{M})| \leq \frac{C}{k}.$$

Theorem 5.2 (OPT vs Φ, Lemma 3.2 of [MS16]). *If $\boldsymbol{M} \in \mathbb{R}^{n\times n}$ satisfies*

$$\|\boldsymbol{M}\|_{\infty\to 2} := \max_{||\boldsymbol{v}||_\infty \le 1} ||\boldsymbol{M}\boldsymbol{v}||_2 \le L\sqrt{n},$$

then for any $\beta > 0$, $k \in [n]$, and $\epsilon \in (0,1]$, there exists a constant C, such that

$$\left|\frac{1}{n}\Phi(\beta, k; \boldsymbol{M}) - \frac{1}{n}\mathcal{OPT}_k(\boldsymbol{M})\right| \le 2L\epsilon\sqrt{k} + \frac{k}{\beta}\log\frac{C}{\epsilon}.$$

Remark 5.3. We can optimize the bound above over ϵ to obtain that when $\epsilon^* = \sqrt{k}/2\beta L \in (0,1]$, we have

$$\left|\frac{1}{n}\Phi(\beta, k; \boldsymbol{M}) - \frac{1}{n}\mathcal{OPT}_k(\boldsymbol{M})\right| \le \frac{k}{\beta}\left(1 + \log\left(\frac{2C\beta L}{\sqrt{k}}\right)\right) = \frac{k}{\beta}\log\frac{C\beta L}{\sqrt{k}}.$$

Remark 5.4. In particular, we can show that (Lemma A.3 of [MS16]) for any graph G with bounded degree, we have $\|\boldsymbol{A}_G^{\text{cen}}\|_{\infty\to 2} \le Cd\sqrt{n}$, and for any $\lambda \in \mathbb{R}$, we have $\|\boldsymbol{B}(\lambda)\|_{\infty\to 2} \le (C + |\lambda|)\sqrt{n}$. so the above bound applies to $\boldsymbol{A}_G^{\text{cen}}$ and $\boldsymbol{B}(\lambda)$ with $L = Cd$ and $L = C + |\lambda|$, respectively.

They further showed the following concentration result for Φ.

Theorem 5.3 (Concentration of Φ, Lemma A.1 of [MS16]). *If $\boldsymbol{M} \in \mathbb{R}^{n\times n}$ is $\boldsymbol{A}_G^{\text{cen}}$ for some graph G with bounded degree, or $\boldsymbol{B}(\lambda)$ for some $\lambda \in \mathbb{R}$, then for any $\beta > 0$ and $k \le n$, for all $t > 0$, we have*

$$\mathbb{P}\left(\left|\frac{1}{n}\Phi(\beta, k; \boldsymbol{M}) - \mathbb{E}\left(\frac{1}{n}\Phi(\beta, k; \boldsymbol{M})\right)\right| \ge t\right) \le Ce^{-nt^2/C}.$$

Therefore, the problem of approximating $\mathcal{SDP}(\boldsymbol{A}_G^{\text{cen}})$ with $\mathcal{SDP}(\boldsymbol{B}(\lambda))$ can be reduced to the problem of interpolating between $\mathbb{E}(\Phi(\beta, k; \boldsymbol{A}_G^{\text{cen}}))$ and $\mathbb{E}(\Phi(\beta, k; \boldsymbol{B}(\lambda)))$. We will defer the details of the interpolation step to the next section, where we examine a more general family of random graph models, the perturbed SBMs. As it turns out, the interpolation step will be the most central step in extending the algorithm to the new setting.

5.2 Lindeberg Interpolation

The motivation of the interpolation step is the following Lindeberg interpolation lemma. cite

Lemma 5.1 (Lindeberg interpolation). *Let $F : \mathbb{R}^N \to \mathbb{R}$ be a thrice continuously differentiable function, and let $\boldsymbol{X} = (X_1, \ldots, X_N)$ and $\boldsymbol{Z} = (Z_1, \ldots, Z_N)$ be two vectors of independent random variables satisfying $\mathbb{E}(X_i) = \mathbb{E}(Z_i)$ and $\mathbb{E}(X_i^2) = \mathbb{E}(Z_i^2)$ for all $i \in [N]$. Then, we have*

$$|\mathbb{E}\left(F(\boldsymbol{X}) - F(\boldsymbol{Z})\right)| \le \frac{1}{6}\left(\max_{i\in[N]}\|\partial_i^3 F\|_\infty\right)\left(\sum_{i=1}^N \mathbb{E}\left[|X_i|^3 + |Z_i|^3\right]\right).$$

Thus, we construct a matrix $\boldsymbol{D}$, such that $\boldsymbol{D}$ is a Gaussian matrix with first and second moments matching

those of $\boldsymbol{A}_G^{\text{cen}}/\sqrt{d}$. Then, apply the above lemma to $F(\boldsymbol{M}) := \Phi(\beta, k; \boldsymbol{M})$ with $\boldsymbol{X} = \boldsymbol{A}_G^{\text{cen}}/\sqrt{d}$ and $\boldsymbol{Z} = \boldsymbol{D}$, we can bound the difference between $F(\boldsymbol{A}_G^{\text{cen}}/\sqrt{d})$ and $F(\boldsymbol{D})$.

First, we bound the third moments of the entries of $\boldsymbol{A}_G^{\text{cen}}/\sqrt{d}$. Note that if $X \sim \text{Ber}(p)$, then for $a \in [0,1]$, $\mathbb{E}[|X-a|^3] = (1-p)a^3 + p(1-a)^3 \le a + p$. Since $\boldsymbol{A}_{G_{ij}}^{\text{cen}} = \boldsymbol{A}_{G_{ij}} - d/n$, and $\boldsymbol{A}_{G_{ij}} \sim \text{Ber}\left(\frac{d}{n}(1+\eta_{ij})\right)$, we have

$$\sum_{i,j\in[n]} \mathbb{E}\left|\frac{1}{\sqrt{d}}\boldsymbol{A}_{G_{ij}}^{\text{cen}}\right|^3 \le \frac{1}{d^{3/2}} \sum_{i,j\in[n]} \frac{d}{n}(2+\eta_{ij}) \le \frac{2n}{\sqrt{d}} + \frac{1}{n\sqrt{d}} \sum_{i,j\in[n]} h_{ij} < \frac{3n}{\sqrt{d}}.$$

finish the argument above

Next, we bound the third moments of the entries of $\boldsymbol{D}$. Note that if $X = a + bZ$, where $Z \sim \mathcal{N}(0,1)$, then $\mathbb{E}[|X|^3] \le 4(|a|^3 + |b|^3)$. Since $\boldsymbol{D}_{ij} = -p_{ij}/\sqrt{d} + \sqrt{p_{ij}(1-p_{ij})/d}\, Z_{ij}$, where $Z_{ij} \sim \mathcal{N}(0,1)$, we have

$$\begin{aligned}
\sum_{i,j\in[n]} \mathbb{E}\left|\boldsymbol{D}_{ij}\right|^3 &\le 4 \sum_{i,j\in[n]} \left(\left(\frac{p_{ij}}{\sqrt{d}}\right)^3 + \left(\frac{p_{ij}(1-p_{ij})}{d}\right)^{3/2}\right) \\
&\le \sum_{i,j\in[n]} \left(\frac{d^{3/2}}{n^3}(1+\eta_{ij})^3 + \frac{1}{n^{3/2}}(1+\eta_{ij})^{3/2}\right) \\
&\le \frac{8d^{3/2}}{n} + 2\sqrt{2n} < \frac{n}{\sqrt{d}},
\end{aligned}$$

where we used the fact that $|\eta_{ij}| \le 1$ and $d \ll n$.

clean up argument

Finally, Montanari and Sen proved that $\|\partial_{ij}^3 F\|_\infty \le 6\beta^2$ for all $i, j \in [n]$ (line 112 of [MS16]). Combining, we obtain

$$\left|\mathbb{E}\left(F(\boldsymbol{A}_G^{\text{cen}}/\sqrt{d}) - F(\boldsymbol{D})\right)\right| \le \frac{Cn\beta^2}{\sqrt{d}}$$

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5.3 Coupling of Gaussian Matrices

For simplicity, we write $\boldsymbol{B}(\lambda) = \boldsymbol{B}$. To bound the difference between $F(\boldsymbol{D})$ and $F(\boldsymbol{B}(\lambda))$, we first notice that, for all matrices $\boldsymbol{M}, \boldsymbol{M}' \in \mathbb{R}^{n\times n}$, we have

$$\begin{aligned}
F(\boldsymbol{M}) &= \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\sum_{i,j=1}^n \boldsymbol{M}_{ij}\langle\boldsymbol{\sigma}_i,\boldsymbol{\sigma}_j\rangle\right) d\nu(\boldsymbol{\sigma})\right)\\
&= \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\sum_{i,j=1}^n \boldsymbol{M}'_{ij}\langle\boldsymbol{\sigma}_i,\boldsymbol{\sigma}_j\rangle + \beta\sum_{i,j=1}^n (\boldsymbol{M}_{ij}-\boldsymbol{M}'_{ij})\langle\boldsymbol{\sigma}_i,\boldsymbol{\sigma}_j\rangle\right) d\nu(\boldsymbol{\sigma})\right)\\
&= F(\boldsymbol{M}') + \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\sum_{i,j=1}^n (\boldsymbol{M}_{ij}-\boldsymbol{M}'_{ij})\langle\boldsymbol{\sigma}_i,\boldsymbol{\sigma}_j\rangle\right) d\nu(\boldsymbol{\sigma})\right)\\
&= F(\boldsymbol{M}') + \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\operatorname{Tr}(\boldsymbol{\Sigma}(\boldsymbol{M}-\boldsymbol{M}')\boldsymbol{\Sigma}^\top)\right) d\nu(\boldsymbol{\sigma})\right)\\
&\le F(\boldsymbol{M}') + \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\|\boldsymbol{M}-\boldsymbol{M}'\|_{\mathrm{op}}\operatorname{Tr}(\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma})\right) d\nu(\boldsymbol{\sigma})\right)\\
&= F(\boldsymbol{M}') + \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(n\beta\|\boldsymbol{M}-\boldsymbol{M}'\|_{\mathrm{op}}\right) d\nu(\boldsymbol{\sigma})\right)\\
&= F(\boldsymbol{M}') + n\|\boldsymbol{M}-\boldsymbol{M}'\|_{\mathrm{op}}.
\end{aligned}$$

where we wrote $\boldsymbol{\Sigma} := (\boldsymbol{\sigma}_1,\dots,\boldsymbol{\sigma}_n) \in (\mathbb{S}^{k-1})^n$, and used the fact that for $\boldsymbol{A}$ symmetric and $\boldsymbol{B} \succeq 0$, since $\boldsymbol{A} \preceq \|\boldsymbol{A}\|_{\mathrm{op}}\boldsymbol{I}$, we have

$$\operatorname{Tr}(\boldsymbol{AB}) \le \operatorname{Tr}(\|\boldsymbol{A}\|_{\mathrm{op}}\boldsymbol{IB}) \le \|\boldsymbol{A}\|_{\mathrm{op}}\operatorname{Tr}(\boldsymbol{B}).$$

Hence,

$$\left|\frac{1}{n}F(\boldsymbol{M}) - \frac{1}{n}F(\boldsymbol{M}')\right| \le \|\boldsymbol{M}-\boldsymbol{M}'\|_{\mathrm{op}}.$$

Therefore,

$$\begin{aligned}
\frac{1}{n}\left|\mathbb{E}F(\boldsymbol{D}) - \mathbb{E}F(\boldsymbol{B})\right| &\le \frac{1}{n}\mathbb{E}\left|F(\boldsymbol{D}) - F(\boldsymbol{B})\right|\\
&\le \mathbb{E}\|\boldsymbol{D}-\boldsymbol{B}\|_{\mathrm{op}}.
\end{aligned}$$

Thus, we want to find a bound on $\mathbb{E}\|\boldsymbol{D}-\boldsymbol{B}\|_{\mathrm{op}}$. To do this, we will use a coupling argument. Writing $V_{ij} = np_{ij}(1-p_{ij})/d$, note that

$$V_{ij} = (1+\eta_{ij})\left(1 - \frac{d}{n}(1+\eta_{ij})\right),$$

$$\boldsymbol{D}_{ij} \sim \mathcal{N}\left(\frac{\lambda}{n}x_ix_j + \frac{\sqrt{d}}{n}h_{ij},\ \frac{1}{n}V_{ij}\right),$$

check signs

$$\boldsymbol{B}_{ij} \sim \mathcal{N}\left(\frac{\lambda}{n} x_i x_j,\ \frac{1}{n}\right).$$

Thus, we can couple $\boldsymbol{D}$ and $\boldsymbol{B}$ such that

diagonal

$$\boldsymbol{D} = \frac{\lambda}{n}\boldsymbol{x}\boldsymbol{x}^\top + \frac{\sqrt{d}}{n}\boldsymbol{H} + \boldsymbol{Z_0} + \boldsymbol{Z_D},$$

$$\boldsymbol{B} = \frac{\lambda}{n}\boldsymbol{x}\boldsymbol{x}^\top + \boldsymbol{Z_0} + \boldsymbol{Z_B},$$

where $\boldsymbol{Z_0}, \boldsymbol{Z_D}, \boldsymbol{Z_B}$ are Gaussian matrices with mean-zero, independent entries. Writing $V_- = \min_{i,j} V_{ij}$, the variances of the entries are given by

$$\mathrm{Var}(\boldsymbol{Z}_{\boldsymbol{0},ij}) = \begin{cases} \frac{1}{n}V_- & \text{if } i \neq j, \\ 0 & \text{if } i = j, \end{cases}$$

$$\mathrm{Var}(\boldsymbol{Z}_{\boldsymbol{D},ij}) = \begin{cases} \frac{1}{n}(V_{ij} - V_-) & \text{if } i \neq j, \\ 0 & \text{if } i = j, \end{cases}$$

$$\mathrm{Var}(\boldsymbol{Z}_{\boldsymbol{B},ij}) = \begin{cases} \frac{1}{n}(1 - V_-) & \text{if } i \neq j, \\ 0 & \text{if } i = j, \end{cases}$$

Under this coupling, by the triangle inequality, we have

$$\mathbb{E}\|\boldsymbol{D} - \boldsymbol{B}\|_{\mathrm{op}} = \mathbb{E}\left\|\frac{\sqrt{d}}{n}\boldsymbol{H} + \boldsymbol{Z_D} - \boldsymbol{Z_B}\right\|_{\mathrm{op}} \leq \frac{\sqrt{d}}{n}\mathbb{E}\|\boldsymbol{H}\|_{\mathrm{op}} + \mathbb{E}\|\boldsymbol{Z_D}\|_{\mathrm{op}} + \mathbb{E}\|\boldsymbol{Z_B}\|_{\mathrm{op}}.$$

We then use the following result for operator norm of centered Gaussian matrices.

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mention symmetry of everything

theorem statement gives op bound of H

cite, 1408.6185

edit V+- defn

Theorem 5.4 (Operator norm of Gaussian matrices)**.** *Let $\boldsymbol{X}$ be an $n \times n$ symmetric matrix with $\boldsymbol{X}_{ij} = g_{ij} b_{ij}$ for $i \geq j$, where g_{ij} are independent standard Gaussian variables, and b_{ij} are given scalars. Then, writing*

$$\sigma^*_{col} = \max_{j \in [n]} \sqrt{\sum_{i=1}^{n} b_{ij}^2}, \quad \sigma^*_{elt} = \max_{i,j \in [n]} |b_{ij}|,$$

for every $s \in (0, 1/2]$, we have

$$\mathbb{E}\|\boldsymbol{X}\| \leq (1+s)\left(2\sigma^*_{row} + \frac{6}{\sqrt{\log(1+s)}}\sigma^*_{elt}\sqrt{\log n}\right)$$

In our case, writing $V_+ = \max_{i,j} V_{ij}$, for all $i, j \in [n]$, we have

$$\mathrm{Var}(\boldsymbol{Z}_{\boldsymbol{D},ij}),\ \mathrm{Var}(\boldsymbol{Z}_{\boldsymbol{B},ij}) \leq \frac{1}{n}(V_+ - V_-).$$

Thus, writing $\delta V := V_+ - V_-$, we have $\sigma^*_{\text{col}} \le \sqrt{\delta V}$, $\sigma^*_{\text{elt}} \le \sqrt{\delta V/n}$. Applying the above theorem,

$$\mathbb{E}\|\boldsymbol{Z_D}\|,\ \mathbb{E}\|\boldsymbol{Z_B}\| \le (1+s)\left(2 + \frac{6}{\sqrt{\log(1+s)}}\sqrt{\frac{\log n}{n}}\right)\sqrt{\delta V} \le C\sqrt{\delta V},$$

when n is sufficiently large.

Recall that $V_{ij} = np_{ij}(1-p_{ij})/d$, and $p_{ij} \ll 1/2$, so V_{ij} is increasing in p_{ij}, and thus increasing in η_{ij}. Since $|\eta_{ij}| \le \epsilon/2 + \|h\|_\infty$, we have

$$\begin{aligned}
\delta V = V_+ - V_- &\le \left(1 + \frac{\epsilon}{2} + \|h\|_\infty\right)\left(1 - \frac{d}{n}\left(1 + \frac{\epsilon}{2} + \|h\|_\infty\right)\right) \\
&\quad - \left(1 - \frac{\epsilon}{2} - \|h\|_\infty\right)\left(1 - \frac{d}{n}\left(1 - \frac{\epsilon}{2} - \|h\|_\infty\right)\right) \\
&\le \left(1 - \frac{d}{n}\right)(\epsilon + 2\|h\|_\infty) \\
&< \epsilon + 2\|h\|_\infty < Cd^{-2\alpha}.
\end{aligned}$$

Therefore, we have

mention eps order in d, or lambda is roughly order 1

$$\begin{aligned}
\mathbb{E}\|\boldsymbol{D} - \boldsymbol{B}\|_{\text{op}} &\le \frac{\sqrt{d}}{n}\mathbb{E}\|\boldsymbol{H}\|_{\text{op}} + \mathbb{E}\|\boldsymbol{Z_D}\|_{\text{op}} + \mathbb{E}\|\boldsymbol{Z_B}\|_{\text{op}} \\
&\le \frac{\sqrt{d}}{n}\left(Cd^{-1/2-\alpha}\right) + C\sqrt{\delta V} \\
&= Cd^{-\alpha}.
\end{aligned}$$

Combining with the bound from the Lindeberg interpolation step, we have

$$\frac{1}{n}\left|\mathbb{E}F\left(\frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \mathbb{E}F(\boldsymbol{B})\right| \le C\left(\frac{\beta^2}{\sqrt{d}} + d^{-\alpha}\right).$$

5.4 Combining all the pieces

With the above bound between $\mathbb{E}F$ values, we first use the concentration of Φ from Theorem 5.3 to show that taking $t = O(d^{-1/2})$, with probability at least $1 - Ce^{-n/C}$, we have

$$\frac{1}{n}\left|\Phi\left(\beta, k; \frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \Phi(\beta, k; \boldsymbol{B})\right| \le C\left(\frac{\beta^2}{\sqrt{d}} + d^{-\alpha}\right).$$

Then, Theorem 5.2 allows us to relate Φ and $\mathcal{OPT}_k$, which gives

$$\begin{aligned}
\frac{1}{n}\left|\mathcal{OPT}_k\left(\frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \mathcal{OPT}_k(\boldsymbol{B})\right| &\le \frac{k}{\beta}\log\frac{C\beta d}{\sqrt{k}} + \frac{k}{\beta}\log\frac{C\beta\lambda}{\sqrt{k}} + C\left(\frac{\beta^2}{\sqrt{d}} + d^{-\alpha}\right) \\
&\le \frac{k}{\beta}\log\frac{C\beta d\lambda}{k} + C\frac{\beta^2}{\sqrt{d}} + Cd^{-\alpha}.
\end{aligned}$$

Note that this holds for all $\beta \in \mathbb{R}$, so taking $\beta = d^{1/6}k^{1/3}$, we have

$$\begin{aligned}\frac{1}{n}\left|\mathcal{OPT}_k\left(\frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \mathcal{OPT}_k(\boldsymbol{B})\right| &\leq Cd^{-1/6}k^{2/3}\log(Cd^{7/6}k^{-2/3}\lambda) + Cd^{-\alpha} \\ &\leq Cd^{-1/6}k^{2/3}\log(Cd^{7/6}\lambda) + Cd^{-\alpha},\end{aligned}$$

where we dropped the $k^{-2/3}$ term in the log since k is a positive integer.

Finally, Theorem 5.1 links $\mathcal{OPT}_k$ and $\mathcal{SDP}$, which gives

$$\frac{1}{n}\left|\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \mathcal{SDP}(\boldsymbol{B})\right| \leq \frac{C}{k} + Cd^{-1/6}k^{2/3}\log(Cd^{7/6}\lambda) + Cd^{-\alpha}.$$

Taking $k = d^{1/10}$, we have

$$\frac{1}{n}\left|\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \mathcal{SDP}(\boldsymbol{B})\right| \leq C\frac{\log(Cd^{7/6}\lambda)}{d^{1/10}} + Cd^{-\alpha}.$$

Crucially, the above bound goes to zero as $d \to \infty$, which allows us to conclude that the same algorithm still works for the perturbed SBM, proving Theorem 4.1. ■

6 Conclusion and Future Directions

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